Circuits & components

Use measurements

1 Charge carries energy

Q = It; E = QV. C = A × s; V = J/C. Conventional current: + to − outside supply.

Metal electrons move oppositely. A cell provides pd; it does not create charge.

2 Resistance

R = V/I in Ω. At fixed V, greater R gives smaller I.

Ammeter in series, voltmeter across component. Use standard symbols; a dot marks a junction.

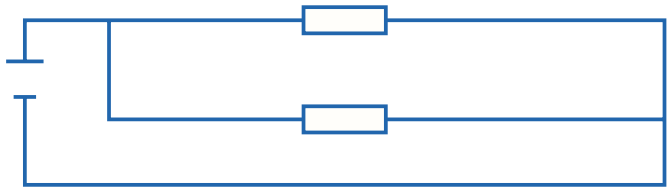
Compare connections

3 One loop: series

Same current throughout. Supply pd = sum of component pds.

4 Ω + 8 Ω across 12 V → 12 Ω total, 1 A; voltage drops 4 V and 8 V.

4 Branches: parallel



Parallel paths share the same pd

Same pd across supply-connected branches; total I = sum of branch currents.

Extra paths reduce total R. One broken branch need not stop others. Appliances can switch independently.

Interpret graphs

5 Resistor & lamp

I vertical, V horizontal: ohmic line through origin at constant temperature; slope = 1/R.

Lamp heats as current rises → ions vibrate more → R increases; curve flattens. Use V/I at a point.

6 Diode & sensors

Diode: high reverse R; conducts forward with sufficient pd. LED emits light.

Thermistor R decreases with temperature; thermostat. LDR R decreases with light; dark-triggered lamp controls.